

Vighnesh JK

Full Stack Developer — AI Engineer

Kerala, India — Open to Relocation — Open to International Opportunities

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Professional Summary

Results-driven Full Stack Developer and AI Engineer with hands-on experience building scalable web applications, AI-powered systems, REST APIs, and intelligent automation platforms. Strong expertise in Python, Django, FastAPI, React, and Generative AI technologies including RAG systems, LangChain, OpenCV, TensorFlow, and Machine Learning frameworks. Skilled in backend architecture, API integration, AI workflow automation, and responsive frontend development. Passionate about building real-world AI solutions and seeking international software engineering opportunities across Europe and Luxembourg.

Technical Skills

Programming Languages: Python, JavaScript, SQL, R

Frontend Technologies: React.js, HTML5, CSS3, Tailwind CSS, Framer Motion, Unity

Backend Technologies: Django, FastAPI, Flask, Node.js, Express.js, Django REST Framework

AI / Machine Learning: TensorFlow, PyTorch, OpenCV, MediaPipe, LangChain, FAISS, XGBoost, Generative AI

Databases: PostgreSQL, MySQL, MongoDB

Tools & Platforms: Git, GitHub, Docker, Linux, Postman, VS Code, Vercel, Render

Core Concepts: REST APIs, JWT Authentication, RBAC, CI/CD, RAG Systems, API Integration, Testing, Debugging, Optimization

Professional Experience

Junior Python Developer

Stroxx Technologies

2025 – Present

Developed scalable backend systems and REST APIs using Django and FastAPI. Designed secure API endpoints and integrated frontend systems. Implemented JWT authentication and role-based access control. Optimized backend performance through debugging and query tuning. Collaborated with agile development teams and participated in deployment, testing, and production support workflows.

Full Stack Developer Intern

SMEC Labs

Built full-stack web applications using Django and frontend technologies. Developed REST APIs and managed backend database operations. Designed responsive user interfaces and optimized application performance for scalable web applications.

Company Projects

Akela Ecommerce – Spec Store Platform

GitHub Repository

- Developed responsive ecommerce platform using React and Tailwind CSS.
- Implemented product browsing, filtering, and API integration workflows.
- Improved frontend responsiveness and optimized user experience.

404 Car Care – Service Management Platform

GitHub Repository

- Built full-stack service management platform using Django and PostgreSQL.

- Developed REST APIs for booking, tracking, and admin workflows.
- Optimized backend architecture and dashboard management systems.

Projects

F.R.I.D.A.Y. AI Assistant

GitHub Repository

- Developed cinematic AI assistant platform inspired by futuristic HUD systems using React, Vite, and Framer Motion.
- Integrated Google Gemini API with Node.js and Express backend for intelligent conversational workflows.
- Designed scalable modular AI architecture with real-time diagnostics simulation and interactive UI feedback.

DocMind – AI Document Search (RAG Platform)

GitHub Repository

- Developed AI-powered RAG platform for conversational querying of PDF documents.
- Implemented semantic search using LangChain and FAISS for accurate information retrieval.
- Built FastAPI backend with React frontend and Docker-based deployment workflows.

AI Agent v4.2.0

GitHub Repository

- Developed intelligent AI automation workflows using LangChain and API integrations.
- Designed modular AI pipeline architecture for scalable task execution systems.
- Implemented autonomous decision-making logic for intelligent workflow automation.

Blind Object Detection System

GitHub Repository

- Developed assistive AI-based object detection system using OpenCV and Unity Engine.
- Implemented real-time object detection and live camera feed analysis for obstacle recognition.
- Designed accessibility-focused audio alert system for visually impaired users.

Sign Language Detection System

GitHub Repository

- Built AI-powered gesture recognition system using OpenCV and MediaPipe.
- Implemented real-time hand tracking and gesture classification workflows.

CloudShare – File Sharing Platform

GitHub Repository

- Built secure backend-integrated file-sharing platform with scalable upload workflows.
- Developed authentication systems and secure file management functionalities.

Certifications

- Python Full Stack Programming Certification

Education

Bachelor of Computer Applications (BCA)

2022 – 2025

STAS College, Pullarikunnu

Soft Skills

Problem Solving, Analytical Thinking, Communication, Teamwork, Leadership, Adaptability, Time Management

Languages

English, Malayalam, Hindi